

6- bargaining

outstanding

- 1.a) The bargaining set
 $U = \{(u_A(x), u_B(x)) : x \in X\}$
 holds the utilities associated with all feasible allocations.

$u_A = 1000F_A + C_A$
 $u_B = 3000F_B + C_B$
 $F_A + F_B = 100; C_A + C_B = 200$

(measure in £'000s)
 scale down by 1000 x for ease.

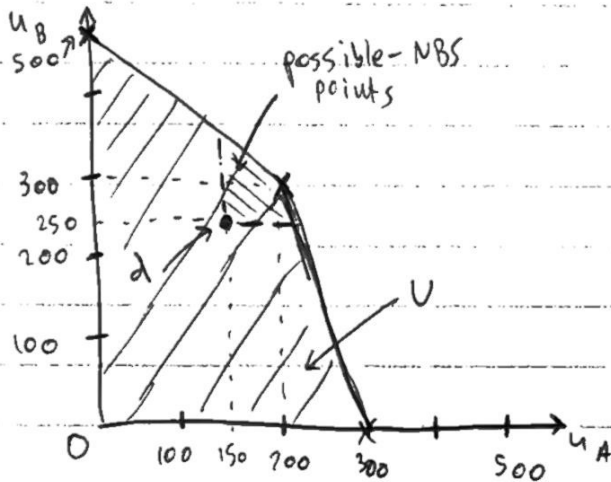
Equal split: $x_d = (50, 100k), (50, 100k)$

$u_A(x_d) = 150, u_B(x_d) = 250$

if A gets all, $u_A = 300$.

B gets all, $u_B = 500$

Farmland to B, cash to A: (200, 300)



- b) we can find NBS by solving

$\max_{u_A, u_B} (u_A - d_A)(u_B - d_B)$ s.t. $(u_A, u_B) \in U$

$u_A \geq d_A, u_B \geq d_B$

- Let's assume that it doesn't bind for either agent, at maximum.
- At the maximum of the Nash product, certainly we'll be on the Pareto frontier of U since otherwise we could've trivially increased it.
- So we have $u_A = F_A + C_A$ with either

(i) $F_A = 0, C_A = 200 - C_B$, or

(ii) $F_A = 100 - F_B, C_A = 200$

Since you only give A farmland once she has all the cash, given B values farmland relatively more.

the

So in (i) the problem is $\max_{C_B, F_B} (200 - C_B - 150)(3F_B + C_B - 250)$

s.t. $F_B = 100, \frac{200}{12} C_B \geq 0$

i.e. $\max_{C_B} (50 - C_B)(50 + C_B)$, which clearly is solved by $C_B = 0$ since then it's a square.

In (ii) the problem is $\max_{C_B, F_B} (100 - F_B + 200 - 150)(3F_B + C_B - 250)$

s.t. $\frac{100}{2} F_B \geq 0, C_B = 0$

$$\text{i.e. } \max_{F_B} (150 - F_B) (3F_B - 250) \quad \text{s.t. } F_B \geq 0$$

with FOC $\frac{\partial \pi}{\partial F_B} = 150 - 6F_B = 0 \Rightarrow F_B = \frac{150}{6} = 25$. So we need to impose constrained maximum at $F_B = 100$ (as obj. func strictly decreasing away from optimum). But that's just the same allocation as in (i).

Thus NBS-outcome is $((0, 200), (100, 0))$; the NBS is $(200, 300)$.

We can't check SOC's since it's at a discontinuity, but this must be the NBS as on either of the line segments making up U's boundary, the (constrained) optimum is at this kink. Moreover, Nash's Theorem tells us that the unique Sol'n satisfying PAR, INV, SYM and IIA can be found by maximising Nash product s.t. IR as we did.

c) ~~if Ann divides~~: If Ann divides:

- B chooses the bundle b s.t. $u_B(b) \geq u_B(b')$
- A wants to create bundles b, b' solving $\max_{u_A(b), u_A(b')} u_A(b')$ s.t. $u_B(b) \geq u_B(b')$.

and in fact the IC constraint on B will bind, since otherwise A could've moved some more goods from b to b' .

(ignore)

Note that we can write the utility Pareto frontier as

$$\begin{cases} u_B = 500 - u_A & \text{for } u_A \in [0, 200] \\ u_B = 900 - 3u_A & \text{for } u_A \in [200, 300] \end{cases}$$

So we require ~~3F_B + c_B = 3F_{B'} + c_{B'}~~

with $c_B + c_{B'} = 200, F_B + F_{B'} = 100$

$$\Rightarrow 300 - 3F_{B'} + 200 - c_{B'} = 3F_{B'} + c_{B'} \Rightarrow 3F_{B'} + c_{B'} = 250.$$

(Oh, of course, they're both half of his full-allocation utility...)

have now checked up
retelling so gets b's

$$\max_{F_b'} 250 - 2F_b' \quad \text{s.t.} \quad \dots$$



trivially solved with $F_b' = 0$,
but constrained optimum.

So A solves $\max_{F_b', c_b'} F_b' + c_b' \quad \text{s.t.} \quad 3F_b' + c_b' = 250$

$$F_b' \in [0, 100]; c_b' \in [0, 200]$$

By inspection we can see A will rather give F_b to B to satisfy indifference before giving c . And it can be satisfied with just F , i.e. $F_b = \frac{250}{3}$, $c_b = 0$

So 2nd path outcome will be F

So SPE is:

- A divides into $(\frac{50}{3}, 200)$, $(\frac{250}{3}, 0)$
 - B chooses bundle b s.t. $u_B(b) > u_B(b')$ or $u_B(b) = u_B(b')$ and $u_A(b') > u_A(b)$
- and utilities on-path will be $(\frac{216\frac{2}{3}}{3}, 250)$

If Bob divides:

$$F_b + c_b = F_b' + c_b' = 250$$

Both bundles must give A utility $\frac{150}{3}$. So B's solution is $u_B(b) = \frac{150}{3}$.
 $3F_b + c_b$ can be written as $3F_b + 150 - F_b = 2F_b + 150$.
 And trivially this is maximised with $F_b = 200$, and $c_b = 150$.

So SPE is:

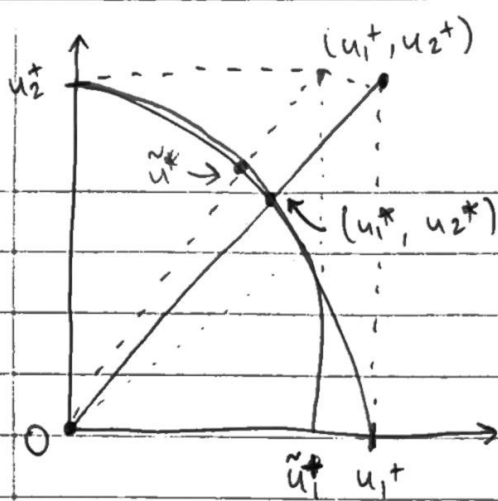
- B divides into $(200, 50)$, $(0, 150)$
 - A chooses similarly to above
- and eq outcome utilities $(150, 250)$

d) we have 50% chance of either outcome. so π payoffs are $(193\frac{1}{3}, 200)$.

Recall the utility frontier from before. We're on the portion $u_A \in [0, 200]$. So at frontier, $u_B = 500 - 193\frac{1}{3} = 316\frac{2}{3} > 200$.
 i.e. we're not on frontier. This is because, although the outcome realized when A divides isn't P-efficient, and likewise with B, a convex combination of them is not, given strictly convex U !

u_i^+ is player i 's best-possible payoff.

2.



a) Graphically: Suppose we restrict U by changing to $\hat{U}$. ~~from~~

We have $f(u, v) \in \hat{U}$ with $\hat{U} \subseteq U$.
But now u_i^+ changed so clearly $(\tilde{u}_1^*, \tilde{u}_2^*) \neq (u_1^*, u_2^*)$.

b) On the frontier, $x_2 = 1 - x_1 = 1 - u_1(x_1)$.
So $u_2(x_2) = (1 - u_1(x_1))^\alpha$ with $\alpha \in (0, 1)$.
and $u_2(1) = 1$.

We have $(u_1^+, u_2^+) = (1, 1)$ by monotonicity. So we're interested in the point where that frontier intersects the line $u_1(x_1) = u_2(x_2)$.

i.e. $u_1(x_1) = (1 - u_1(x_1))^\alpha$
 $(\Rightarrow) x_1 = (1 - x_1)^\alpha$
 $x_1 + x_1^{1/\alpha} = 1$

Suppose $(x_1^*, x_2^*) = (\frac{1}{2}, \frac{1}{2})$.

Then LHS $= \frac{1}{2} + \frac{1}{2}^{1/\alpha}$ for some $\alpha \in (0, 1)$.
 but $\frac{1}{2}^\beta$ for $\beta := 1/\alpha > 1$ is $< \frac{1}{2}$.
 So LHS < 1 , equality fails.

And LHS strictly increasing in x_1 , since
 $\partial/\partial x_1 = 1 + \frac{1}{\alpha} x_1^{1/\alpha - 1} > 0$.

So when satisfied, we must have $x_1^\alpha > \frac{1}{2}$
 $\Rightarrow x_1^* > x_2^*$.

c) Now $(u_1^+, u_2^+) = (s, s)$; frontier traced out by $u_2(x_2) = (s - u_1(x_1))^\alpha$.
 The relevant ray still has slope 45° .

So we want to find the allocation s.t.

$$u(x_1) = (s - u_1(x_1))^\alpha$$

$$(\Rightarrow) x_1 + x_1^{1/\alpha} = s$$

Clearly x_1 's allocation in s increases, by LHS strictly increasing and $s \geq 1$.

And note that by construction

$$\frac{u_1(x_1^*)}{u_1^*} = \frac{u_2^*}{u_2^*}$$

c). Now $(u_1^*, u_2^*) = (s, s^\alpha)$. Frontier traced out by $u_2(x_2) = (s - u_1(x_1))^\alpha$.

We want to find $x_2^* = s - x_1^*$ s.t.

$$\frac{u_1(x_1^*)}{s} = \frac{u_2(x_2^*)}{s^\alpha}$$

i.e.
$$\frac{x_1^*}{s} = \frac{(s - x_1^*)^\alpha}{s^\alpha}$$

$$\Rightarrow x_1^* + \left(\frac{x_1^*}{s}\right)^{1/\alpha} = s.$$

- In the case where $s \geq 1$ we recover previous condition.
- We can see that for $s > 1$, the implicit value of x_1^* pinned down is strictly increasing in s (could use IFT, but by inspection: RHS increases and second term on LHS decreases, fixing x_1^* , so it must $\uparrow$).

Both

Equivalently, since $u_1(x_1^*) = s - x_2^*$ we have

$$\frac{s - x_2^*}{s} = \frac{x_2^{*\alpha}}{s^\alpha} = 1 - \frac{x_2^*}{s}$$

Nice

$$\text{so } \frac{x_2^*}{s} + \left(\frac{x_2^*}{s}\right)^\alpha = 1.$$

The x_2^* this pins down is similarly strictly increasing in s , since LHS denominator's grow, and rest all fixed.

So for $s > 1$, we do have both getting strictly larger allocations. This is in contrast to MBS which is non-monotonic, ~~high-risk areas prefer assets to get assets with~~ ~~more risk in general.~~

3. $0 \leq Y_i \leq X_i \leq 1$.

$F(x) = x^2$

and we'd have $G(y) = y(1-y)$

a) The ~~budget set~~ x_1 (i) (ii) (iii) — regions where x_1 could lie

$X_1 = 0$ w.p. 0 so we can ignore that; likewise any ties.

There are three cases to consider:

(i) $X_1 \leq b \leq a$ w.p. b^2 ; P2 wins both, pays $2X_1$.

(ii) $b \leq X_1 \leq a$ w.p. $a^2 - b^2$; P2 wins one, pays b .

(iii) $b \leq a \leq X_1$ w.p. $1 - a^2$; P2 wins ~~one~~ ^{one}, pays b .

In (iii) payoff for P2 is $X_2 - b$

(ii) payoff is $X_2 - b$

(i) payoff is $X_2 + Y_2 - 2X_1$, but X_1 is unknown to P2,

so we need to find its expectation given

~~$X_1 \leq b$~~ $X_1 \leq b$ here.

So, we have

$$v_2 = \cancel{b^2} (a^2 - b^2 + 1 - a^2) (X_2 - b) + b^2 (X_2 + Y_2 - 2E[X_1 | X_1 \leq b])$$

The first term simplifies to $(1 - b^2)(X_2 - b)$ as required.

The second term is $Pr(X_1 \leq b) \cdot (X_2 + Y_2 - 2E[X_1 | X_1 \leq b])$.

By LOTUS this

what! LOTUS!

$$= \int_0^b (X_2 + Y_2 - 2x) 2x dx \text{ as required.}$$

b) P2 needs ~~max~~ to $\max_{a,b} v_2(\cdot)$, for each of her (continuous) types in playing $\{X_2, 0\}$ for it to be part of BNE.

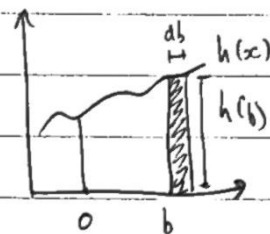
a doesn't appear in the payoffs at all. So we can ignore it, provided that $b \leq a$, and just optimise b .

$$\frac{\partial v_2}{\partial b} = -2b(x_2 - b) - (1 - b^2) + \frac{\partial}{\partial b} \int_0^b (x_2 + y_2 - 2x) 2x dx.$$

where the third term is

$$(x_2 + y_2 - 2b) 2b$$

$$= -b^2 - 1 + 2y_2 b.$$



$$\frac{d}{db} \int_0^b h(x) dx = h(b).$$

$$\Rightarrow \frac{d}{db} \int_0^b h(x) dx = h(b).$$

And second derivative $\frac{\partial^2 v_2}{\partial b^2} = -2b + 2y_2.$

This is decreasing in b , and at the FOC

$$-b^2 - 1 + 2y_2 b = 0$$

$$\Leftrightarrow b^2 - 2y_2 b + 1 = 0$$

For the optimum to be $b=0$, we want a constrained maximum. So the FOC might not bind (the Lagrangian would) and we can just show that the Π payoff is decreasing in b on the interval $[0, 1]$.

$$-b^2 - 1 + 2y_2 b \leq 0 \Leftrightarrow b^2 - 2y_2 b + 1 \geq 0.$$

$$\Leftrightarrow (b - y_2)^2 - y_2^2 + 1 \geq 0$$

$$\begin{array}{c} \uparrow \quad \uparrow \\ \geq 0 \text{ since } \cdot^2 \quad \geq 0 \text{ since } y_2 \in [0, 1]. \end{array}$$

So $b=0$ is indeed optimal BR to $\{x_1, 0\}$.

And since a is immaterial to p_2 's payoffs, let it $= x_2$.

Then we could make a symmetric argument for p_1 , using (a). Thus all types of both players BR to the other, so it's BNE.

c) ~~The two players have different valuations for the second item w.p. 1. But in this setup, it's simply arbitrary (i.e. decided by some tiebreak rule) which of them receives it, since they both bid 0 for it at this particular BNE. (It's possible there are other BNE which would be efficient?).~~

c) In situations like (i), where one player values both items more highly than their opponent values either, we'll get an inefficient allocation — each player gets one item (and pays 0) rather than the individual who values them most receiving both goods.

(It's conceivable we have other, efficient BVE though?)

Intuitively, the problem is that we have a uniform price. If I increase my bid b_i to a truthful amount and win both, I push up the price I'm made to pay for both items.

Great

We can fix this by using a pivotal mechanism of the VCG flavour. For each unit won, the winner ought to pay the highest bid that would've won had they not participated at all, i.e. total price is $\sum_{j \neq i} b_j$ bids that would've won without i . This means agents pay their external effect on others' welfare and incentivises truth-telling by aligning preferences with SWF.